

Correlation Theory and Regression Analysis

1. What is linear regression?

Ans: The probable movement of one variable in terms of other variable is called Linear regression. For example, the yield of paddy depends on water, fertilizer and irrigation etc.

$$y = a + bx$$

$$y = a + b_1X_1 + b_2X_2 + b_3X_3$$

2. Define regression coefficients?

Ans: The mathematical measures of regression are called the coefficient regression.

The regression coefficients y on x

$$b_{xy} = \frac{\sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

The regression coefficients x on y

$$b_{yx} = \frac{\sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

3. properties of regression coefficient.

Ans:

- i. Regression coefficient lies between $-\infty$ to $+\infty$.
- ii. Regression coefficient is not symmetric. i.e, $b_{xy} \neq b_{yx}$
- iii. The geometric mean of regression coefficients is equal to correlation coefficient

$$r_{xy} = \sqrt{b_{yx} \times b_{xy}}$$
- iv. The arithmetic mean of two regression coefficient is greater than correlation Coefficient.
- v. Regression coefficient is not unit free.

4. State some uses of regression in engineering statistics.

Ans:

- | | |
|-------------------------|---------------------------------------|
| i. Predictive Modeling | vi. Process Improvement: |
| ii. Design Optimization | vii. Risk Assessment: |
| iii. Quality Control | viii. Cost Estimation: |
| iv. Failure Analysis | ix. Environmental Engineering: |
| v. Signal Processing | x. Mechanical and Structural Analysis |

5. What is correlation?

Ans:

If the change in one variable affects a change in the other variable, the variables are said to be correlated.

For example, the production of paddy is dependent on the rainfall. Here production of paddy is considered to be a dependent variable.

6. Define correlation coefficients with example

Ans:

The **correlation coefficient** is a statistical measure that indicates the strength and direction of the linear relationship between two variables. It is denoted by **r** and its value ranges between **-1** and **1**

7. Write down the properties of correlation coefficient.

Ans:

- Correlation coefficient lies between -1 to +1.
- Correlation coefficient is symmetric. i.e, $r_{xy} = r_{yx}$
- Correlation coefficient is the geometric mean of regression coefficients i.e, $r_{xy} = \sqrt{b_{yx} \times b_{xy}}$
- For two independent variable correlation coefficient is zero
- It is always unit free

8. Distinguish between correlation coefficients and regression coefficients.

Regression Coefficients	Correlation Coefficients
a) Regression coefficient lies between $-\infty$ to $+\infty$.	a) Correlation coefficient lies between -1 to +1
b) Regression coefficient is not symmetric. i.e, $b_{xy} \neq b_{yx}$	b) Correlation coefficient is symmetric. i.e, $r_{xy} = r_{yx}$
c) It is always not unit free	c) It is always unit free

9. Interpret the following correlation coefficients:

r value	Interpretation
$r=1$	Perfect positive linear correlation
$0.8 \leq r < 1$	Strong positive linear correlation
$0.4 \leq r < 0.8$	Moderate positive linear correlation
$0 < r < 0.4$	Weak positive linear correlation
$r=0$	No correlation
$-0.4 \leq r < 0$	Weak negative linear correlation
$-0.8 \leq r < -0.4$	Moderate negative linear correlation
$-1 \leq r < -0.8$	Strong negative linear correlation
$r = -1$	Perfect negative linear correlation

10. A researcher computes the correlation coefficient between two variables and gets $r = -1.02$. What does this value mean?

1.02. What does this value mean?

Ans:

If a researcher computes the correlation coefficient $r = -1.02$, this result is **invalid**, because the correlation coefficient must always lie within the range of **-1 to 1**.

Fundamental of Probability

1. Distinguish between Experiment and Random experiment.

Ans:

Experiment

Experiment is an act that can be repeated under given condition. Like tossing of a coins.

Random Experiment

Experiment are called random experiment if the outcomes depend on chance and not be predicted with certainly.

2. Outcomes?

Ans: The result of an experiment are called outcomes.

3. Sample space?

Ans:

Sample space: The collection or totality of all possible outcomes of a random experiment is called sample space. It is usually denoted by S or Ω .

If we toss a coin, the sample space is $S = \{H, T\}$ where H and T denote the head and tail of the coin respectively.

4. Sample point?

Ans: Each element of a sample space is called sample point

5. Event?

Ans: Any subset of a sample space is called event. There are two types of event:

Simple event and Compound event

6. Simple event?

Ans: An event is called simple event if it contains only one sample point.

7. Conditional Probability?

Ans:

If **A** and **B** are two events in **S**. Then the conditional probability of **A** for given value of **B**, denoted by $P[A | B]$ is defined by

$$P[A | B] = \frac{P[AB]}{P[B]} ; P[B] > 0$$

$$\text{Similarly, } P[B | A] = \frac{P[AB]}{P[A]} ; P[A] > 0$$

8. Present illustrative examples to distinguish between independent event and dependent event.

Ans:

Independent Event: Two events A and B are said to be independent if and only if one of the following conditions holds:

- (i) $P[AB] = P[A]P[B]$
- (ii) $P[A | B] = P[A]$
- (iii) $P[B | A] = P[B]$

Dependent Event: Two events A and B are said to be dependent if and only if one of the

Following conditions holds:

- (i) $P[AB] \neq P[A]P[B]$
- (ii) $P[A | B] \neq P[A]$
- (iii) $P[B | A] \neq P[B]$

9. Distinguish between Mutually exclusive events and Non-mutually exclusive events.

Ans:

Mutually exclusive events: Two events are said to be mutually exclusive if they have no common points. If A and B are two mutually exclusive events, then $AB = \emptyset$.

Non mutually exclusive events: Two events are said to be not mutually exclusive event if they have common points. If A and B are two not mutually exclusive events, then $AB \neq \emptyset$

10. Probability?

Ans: If A be the event is S, then the probability of the event A, denoted by $P[A]$ is defined as,

$$P[A] = \frac{\text{Favorable number of Outcomes}}{\text{Total number of Outcomes}}$$

$$P[A] = \frac{N[A]}{N[S]} = \frac{m}{n} \quad ;$$

$$0 \leq P[A] \leq 1$$

11. Axiomatic probability?

Ans: If A be the event is S, Then the probability of the event A, denoted by $P[A]$ must satisfy following four axioms,

- (i) $P[A] \geq 0$
- (ii) $P[S] = 1$
- (iii) If A and B are mutually exclusive events, then $P[A \cup B] = P[A] + P[B]$
- (iv) Let $A_1, A_2, \dots, A_k$ be a sequence of K mutually exclusive events, then $P[A_1 \cup A_2 \cup \dots \cup A_k] = P[A_1] + P[A_2] + \dots + P[A_k]$

Random Variable and Mathematical Expectation

1. Define random variable with example?

Ans:

The random is a real valued function whose values are determined with the outcomes of a random experiment. It is usually denoted by X, Y, Z

Let us consider the experiment of tossing two fair coins. The sample space of the experiment is $S = \{HH, HT, TH, TT\}$

Let X denotes the number of heads. So,

Sample point	Number of Head (X)
HH	2
HT	1
TH	1
TT	0

2. Distinguish between discrete and continuous random variables.

Ans:

Discrete random variable: A random variable is called discrete random variable if it can take only isolated values. Example: Family members, mobile number etc

Continuous random variable: A random variable is called continuous random variable if it can take any values between certain limits. Example: Age, weight, height etc

3. What is Probability function ?

Probability function: A function $f(x)$ of a discrete random variable X is called a probability function if it satisfies the following two conditions:

(i) $f(x) \geq 0$

(ii) $\sum f(x) = 1$

4. What is Probability density function?

Ans:

Probability density function: A function $f(x)$ of a continuous random variable X is called a probability density function if it satisfies the following two conditions:

(i) $E[X] = \sum f(x) dx$

(ii) $E[X] = \int_{-\infty}^{\infty} f(x) dx$

5. Define Mathematical Expectation of a random variable.

Ans:

If X is a discrete or continuous random variable with probability function or probability density function $f(x)$. then the mathematical expectation of X is usually denoted by $E[X]$ or μ and defined by

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} xf(x) = 1$

6. Write down the Properties of mathematical expectation of a random variable.

Ans:

- (i) If $\mathbf{b}$ is a constant then $\mathbf{E[b] = b}$
- (ii) If $\mathbf{X}$ is a random variable with expectation $\mathbf{E[X]}$, then $\mathbf{E[aX+b] = aE[X] + b}$, Where $\mathbf{a}$ and $\mathbf{b}$ constant.
- (iii) If $\mathbf{X}$ is a random variable with expectation $\mathbf{E[X]}$, then $\mathbf{E[X-E(X)] = 0}$
- (iv) If $\mathbf{X}$ and $\mathbf{Y}$ are random variables then $\mathbf{E[X+Y] = E[X] + E[Y]}$
- (v) If $\mathbf{X}$ and $\mathbf{Y}$ are random variables then $\mathbf{E[X-Y] = E[X] - E[Y]}$

7. Write down the properties of the variance of a random variable.

Ans:

- (i) If $\mathbf{b}$ is a constant then $\mathbf{V[b] = 0}$
- (ii) If $\mathbf{X}$ is a random variable, then $\mathbf{V[aX+b] = a^2V[X]}$, Where $\mathbf{a}$ and $\mathbf{b}$ constant.
- (iii) If $\mathbf{X}$ is a random variable with expectation $\mathbf{E[X]}$, then $\mathbf{V(X) = E[X-E(X)]^2 = E(X^2) - [E(X)]^2}$
- (iv) If $\mathbf{X}$ and $\mathbf{Y}$ are random variables then $\mathbf{V[X+Y] = V[X] + V[Y]}$
- (v) If $\mathbf{X}$ and $\mathbf{Y}$ are random variables then $\mathbf{V[X-Y] = V[X] + V[Y]}$

8. What does the expected value of a random variable measure? Could the expected value be negative?

Probability Distributions

1. Introduce the binomial distribution along with its assumptions/ inherent assumptions?.

Ans:

A discrete random variable X is said to have binomial distribution if its probability function is as follows:

$$f(x) = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$$

Where **n** and **p** are the parameters of the distribution and **p + q = 1**

We get the binomial distribution under the following experimentation conditions

1. The number of trial n is finite
2. The trials are independent of each other
3. The probability of success p is constant for each trial.
4. Each trial must result in a success or failure.
5. The events are discrete events.

2. Under what conditions will binomial distribution tend to Poisson distribution?

3. State the conditions under which a binomial distribution can be approximated as a normal distribution. ?

4. Define Poisson distribution?

Ans:

A discrete random variable X is said to have Poisson distribution if its probability function is as follows:

$$f(x) = \frac{e^{-m} m^x}{x!}$$

Where m is the parameter of the distribution and $e = 2.718$

5. Define normal probability distribution with its importance.

Ans:

A continuous random variable X is said to have normal distribution if its probability density function is as follows

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \begin{matrix} -\infty \leq x \leq \infty \\ -\infty \leq \mu \leq \infty \\ \sigma^2 \geq 0 \end{matrix}$$

μ and σ^2 are the parameters of the distribution.

Importance of normal distribution:

1. In practice under certain condition most of the probability and sampling distributions can be approximated by normal distribution.
2. According to central limit theorem, if mean and variance of a distribution exist, then the

distribution converted to normal distribution.

3. Normal distribution is the basis of all the sampling distribution.
4. Assumption of normality is the basis of all the test of significance in applied statistics.
5. Normal distributions find its application in industrial statistics such as quality control.

6. Write some practical situations suitable for Poisson distribution.

Ans:

1. It is used in quality control statistics to count the number of defects of an item.
2. In biology, to count the number of bacteria.
3. In determining the number of deaths in a district in a given period, by rare disease.
4. The number of error per page in typed material.
5. The number of plants infected with a particular disease in a plot of field.
6. Number of weeds in particular species in different plots of a field.
7. Discuss the importance of normal distribution in engineering field.

7. What are the parameters of a binomial distribution? Why are they so called?

Estimation and Test of Hypothesis

1. What is Hypothesis?

Ans: A statement about the nature of a population.

Example: Students who eat breakfast will perform better on a statistics exam than students who do not eat breakfast.

2. Null Hypothesis?

Ans: The hypothesis which we are going to test for possible rejection under the assumption that it is true. Null hypothesis is denoted by H_0 ; $H_0: \mu_1 = \mu_2$

3. Alternative Hypothesis?

Ans: Each of all possible hypothesis other than null hypothesis is called alternative hypothesis and is usually denoted by H_1 ; $H_0: \mu_1 \neq \mu_2$

4. Distinguish between the followings Type I error and Type II error

Ans:

Type I : The error of rejecting H_0 (accepting H_1) which is true is called error of first kind or Type-I error.

Type II: The error of accepting null hypothesis H_1 when it is false is called error of 2nd kind or H_1 Type-II error is denoted by β

5. Explain Level of significance ?

Ans: The probability of Type-I error denoted by α is called level of significance. Symbolically $\alpha = P(\text{rejecting } H_0 | H_0 \text{ is true})$. It is also known as size of a test

6. Explain Level of P-Value?

Ans: The p value is the least possible value of alpha for which we can reject the null hypothesis.

- If $p < 0.01$, the results are regarded as highly significant
- If $0.01 < p < 0.05$, the results are regarded as statistically significant
- If $p > 0.10$, the results are considered not significant

7. Commonly used test statistics?

Ans:

- a) The normal test (z-test)
- b) The t test
- c) The χ^2 -test
- d) The F test

8. Write some applications of χ^2 -test from engineering perspective.

Ans:

- a. To test the significance of a specified population variance
- b. To test the goodness of fit of a distribution
- c. To test the independence of attributes
- d. To test the equality of several correlation coefficients
- e. To test the homogeneity of several tests

f. To test the homogeneity of several population variance